In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

In [5]:
```
df = pd.read_csv(r"C:\Users\Srushti\Downloads\netflix_titles.csv\netflix_titles.csv

df.head()
```

Out[5]:

| | show_id | type | title | director | cast | country | date_added | release_year | rati |
|---|---|---|---|---|---|---|---|---|---|
| **0** | s1 | Movie | Dick Johnson Is Dead | Kirsten Johnson | NaN | United States | September 25, 2021 | 2020 | PG- |
| **1** | s2 | TV Show | Blood & Water | NaN | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa | September 24, 2021 | 2021 | T N |
| **2** | s3 | TV Show | Ganglands | Julien Leclercq | Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi... | NaN | September 24, 2021 | 2021 | T N |
| **3** | s4 | TV Show | Jailbirds New Orleans | NaN | NaN | NaN | September 24, 2021 | 2021 | T N |
| **4** | s5 | TV Show | Kota Factory | NaN | Mayur More, Jitendra Kumar, Ranjan Raj, Alam K... | India | September 24, 2021 | 2021 | T N |

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

In [7]:
```
df.columns
```

Out[7]:
```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')
```

In [9]:
```
df['date_added'] = pd.to_datetime(df['date_added'],
errors='coerce')
```

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        98
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

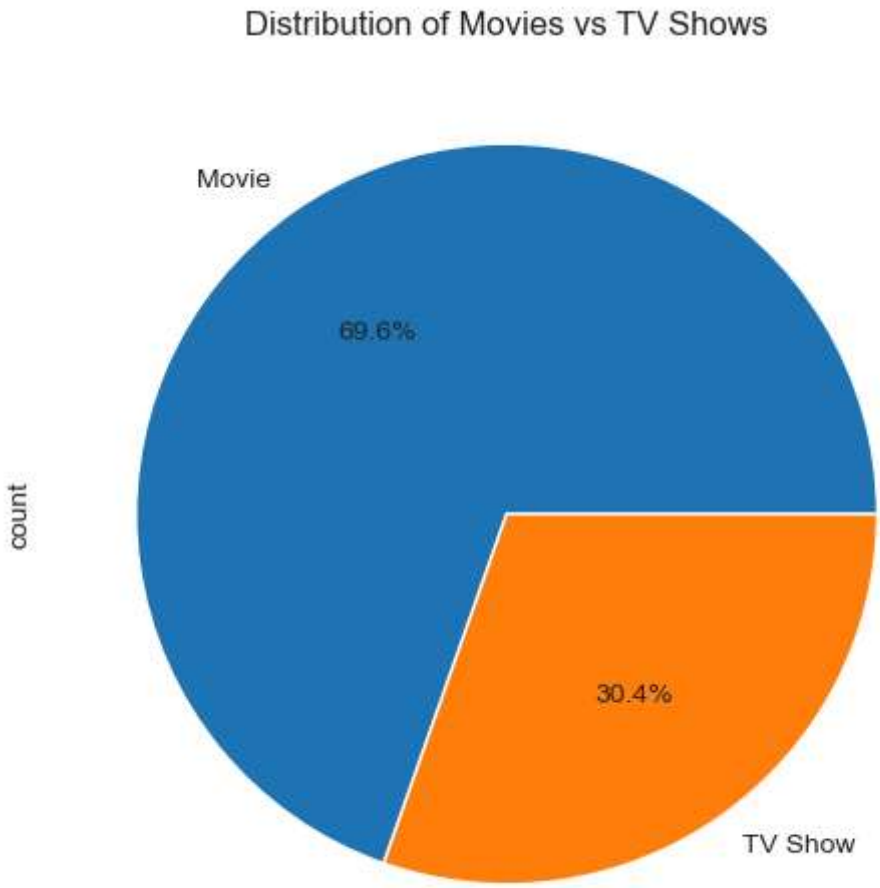
In [12]:
```
df['country'].fillna("Unknown")
df['director'].fillna("Unknown")
```

Out[12]:
```
0       Kirsten Johnson
1               Unknown
2       Julien Leclercq
3               Unknown
4               Unknown
                ...
8802      David Fincher
8803            Unknown
8804    Ruben Fleischer
8805       Peter Hewitt
8806        Mozez Singh
Name: director, Length: 8807, dtype: object
```

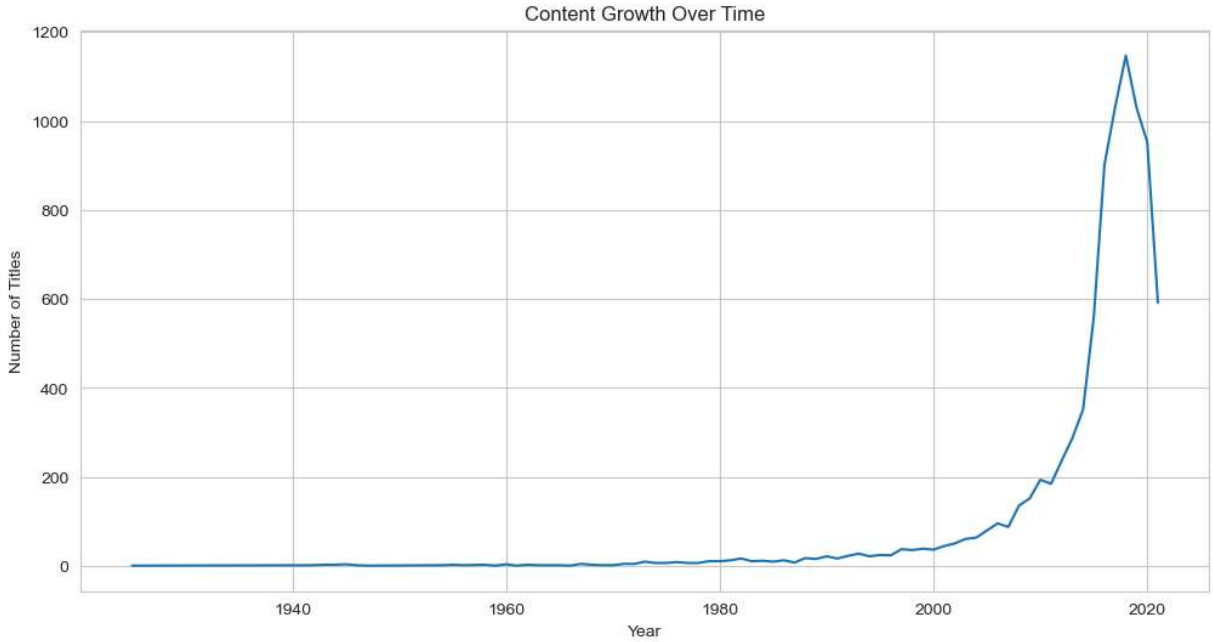
In [13]:
```
type_counts = df['type'].value_counts()

print(type_counts)
```

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
The history saving thread hit an unexpected error (OperationalError('database or dis
k is full')).History will not be written to the database.
```

In [14]:
```
plt.figure(figsize=(6,6))

type_counts.plot(kind='pie', autopct='%1.1f%%')

plt.title("Distribution of Movies vs TV Shows")

plt.savefig("type_distribution.png", dpi=300)

plt.show()
```



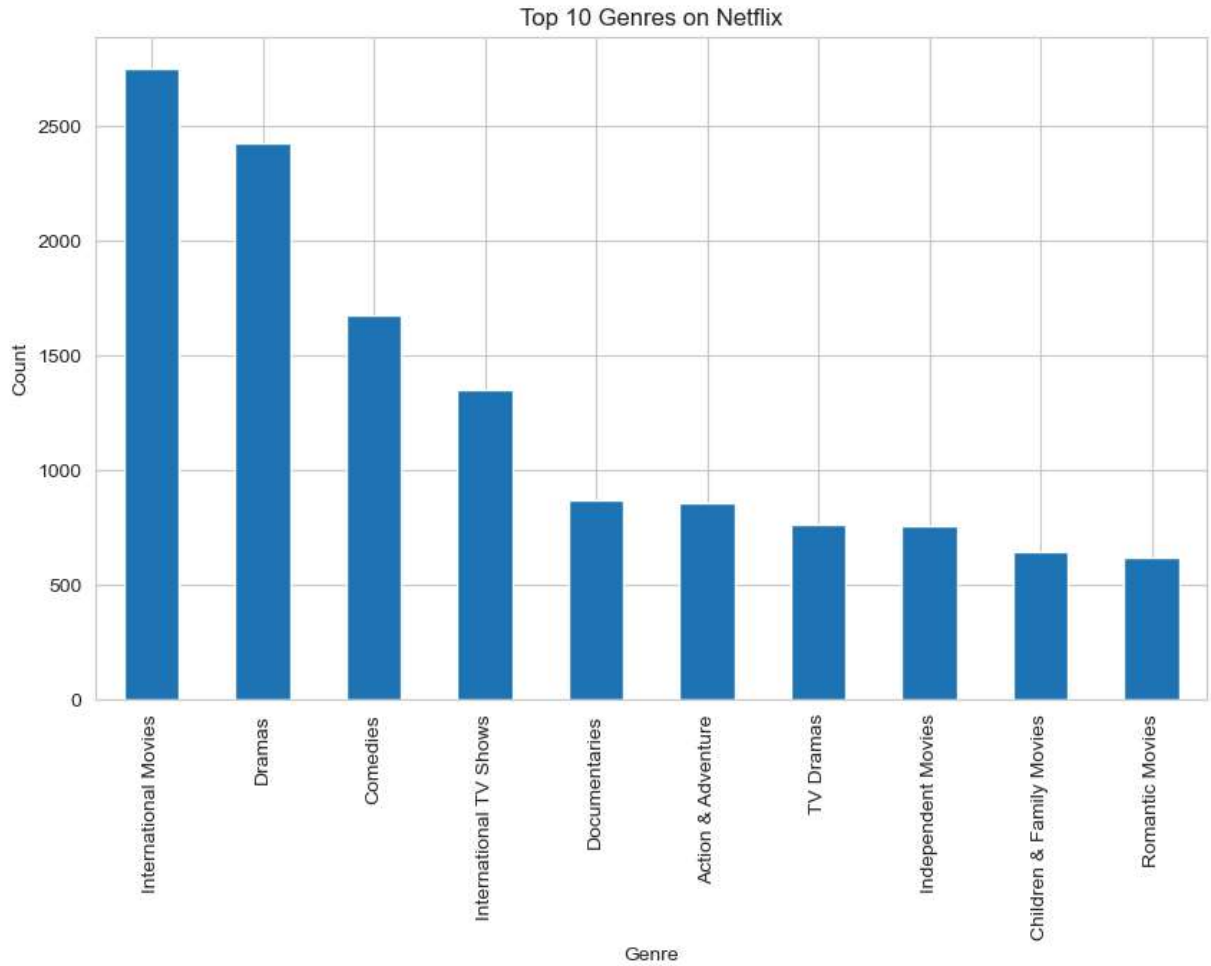

In [18]:
```
year_counts = df['release_year'].value_counts().sort_index()


plt.figure(figsize=(12,6))

year_counts.plot()

plt.title("Content Growth Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Titles")

plt.savefig("content_growth.png", dpi=300)

plt.show()
```




In [19]:
```
genres=df['listed_in'].str.split(', ')
all_genres=[]
for g in genres:
    all_genres.extend(g)
genre_series=pd.Series(all_genres)
top_genres=genre_series.value_counts().head(10)
print(top_genres)
```

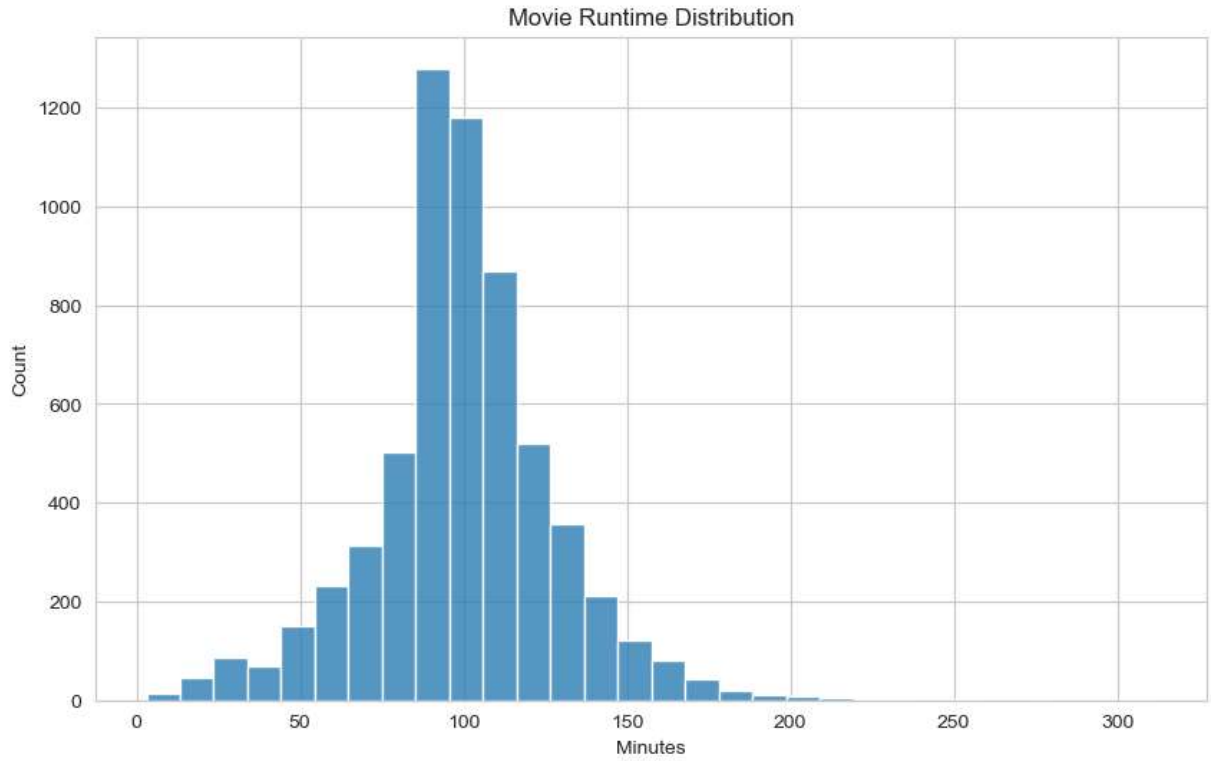
```
International Movies        2752
Dramas                      2427
Comedies                    1674
International TV Shows      1351
Documentaries                869
Action & Adventure           859
TV Dramas                    763
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64
```

In [20]:
```
plt.figure(figsize=(10,6))

top_genres.plot(kind='bar')

plt.title("Top 10 Genres on Netflix")
plt.xlabel("Genre")
plt.ylabel("Count")

plt.savefig("top_genres.png", dpi=300)

plt.show()
```




In [24]:
```
movies = df[df['type'] == "Movie"].copy()

movies['minutes'] = movies['duration'].str.replace(" min","")

movies['minutes'] = pd.to_numeric(movies['minutes'], errors='coerce')
```

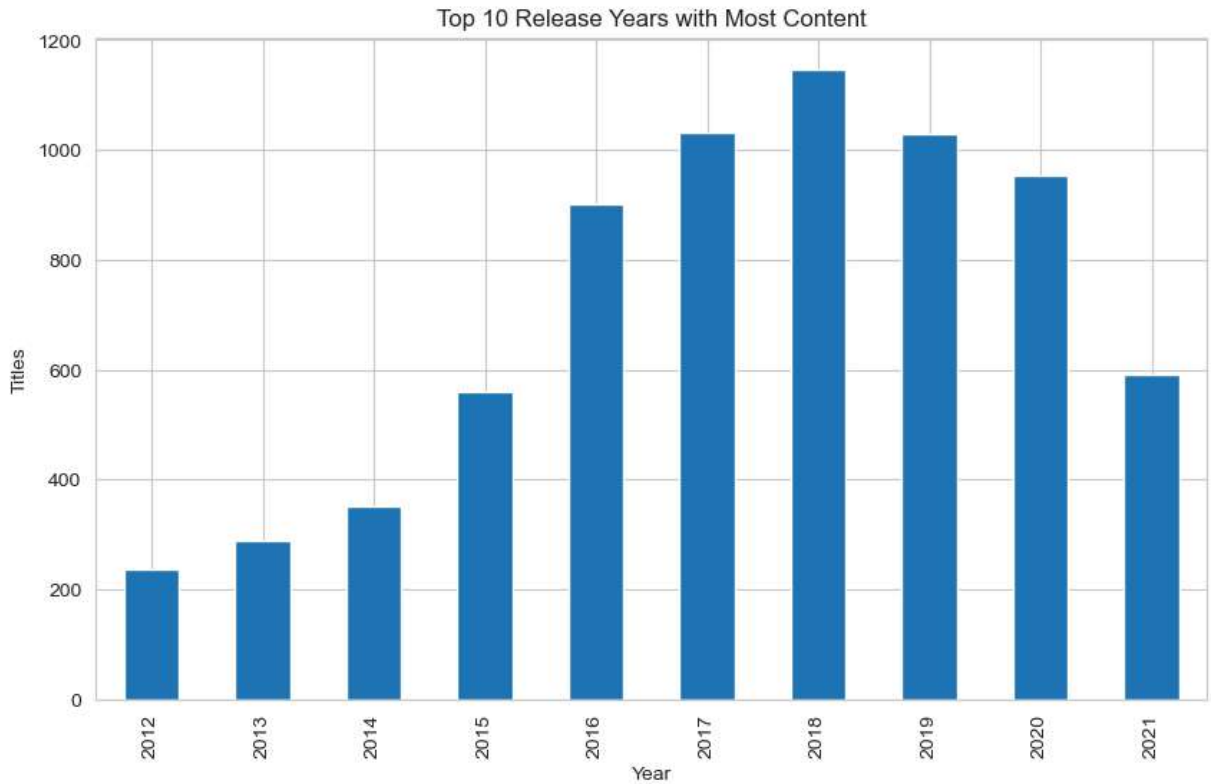
In [25]:
```
plt.figure(figsize=(10,6))

sns.histplot(movies['minutes'], bins=30)

plt.title("Movie Runtime Distribution")
plt.xlabel("Minutes")

plt.savefig("runtime_distribution.png", dpi=300)

plt.show()
```




In [26]:
```
top_years = df['release_year'].value_counts().head(10)

print(top_years)
```

```
release_year
2018    1147
2017    1032
2019    1030
2020     953
2016     902
2021     592
2015     560
2014     352
2013     288
2012     237
Name: count, dtype: int64
```

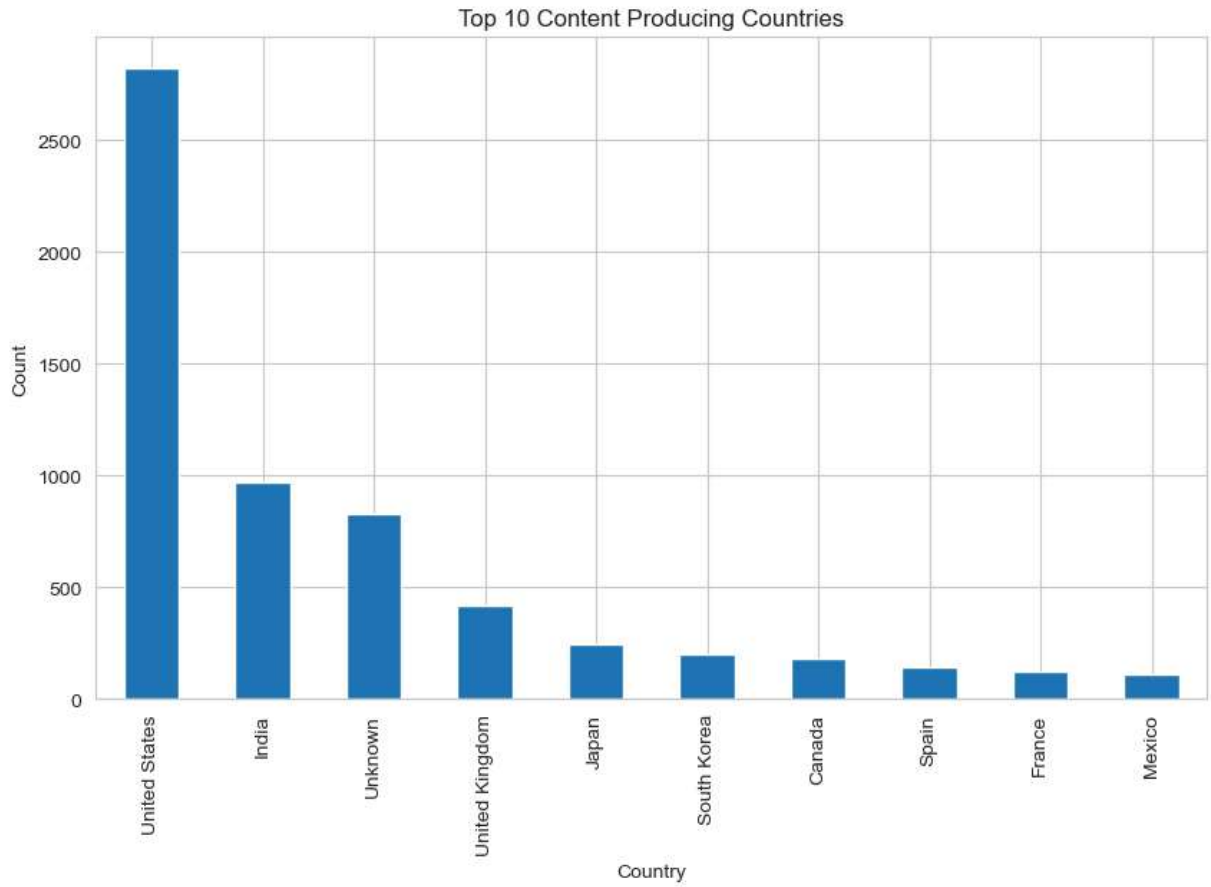
In [27]:
```
plt.figure(figsize=(10,6))
top_years.sort_index().plot(kind='bar')
plt.title("Top 10 Release Years with Most Content")
plt.xlabel("Year")
plt.ylabel("Titles")

plt.savefig("top_years.png",dpi=300)
plt.show()
```




In [28]:
```
top_countries=df['country'].value_counts().head(10)
print(top_countries)
```

```
country
United States     2818
India              972
Unknown            831
United Kingdom     419
Japan              245
South Korea        199
Canada             181
Spain              145
France             124
Mexico             110
Name: count, dtype: int64
```

In [29]:
```
plt.figure(figsize=(10,6))
top_countries.plot(kind='bar')
plt.title("Top 10 Content Producing Countries")
plt.xlabel("Country")
plt.ylabel("Count")
plt.savefig("top_countries.png",dpi=300)
plt.show()
```


Top 10 Content Producing Countries
Count
2500
2000
1500
1000
500
0
United States
India
Unknown
United Kingdom
Japan
South Korea
Canada
Spain
France
Mexico
Country

In [ ]: